

Discussion handout — Khaleel & Al-Nimri 2026: deep bite with bite turbos alone, plus a reverse-curve wire, or plus posterior elastics

Khaleel R, Al-Nimri K. Assessment of three different techniques in correcting deep overbite: a prospective clinical study. Angle Orthod. 2026;96(5):528–535. DOI 10.2319/100925-850.1.

Design: prospective, non-randomised, single-operator, operator-allocated | n: 81 enrolled, 80 completed, 27/27/27 nominal | Groups: I turbos+0.016×0.022 NiTi RCOS; II turbos+0.014 NiTi+box elastics; III turbos+0.014 NiTi | Cephalograms at bonding and after levelling.

30-second opening

"Everyone had bite turbos; does a reverse-curve wire or box elastics speed bite opening? Yes: elastics 44 days, reverse curve 56, turbos alone 94; an auxiliary halves it. The reverse curve cost 7.5° of incisor flaring and 6–9° of molar tip-back; elastics flared 3.5° but opened the mandibular plane most. The operator chose the arms, partly on cooperation, so I take the halving, not the 11-day ranking."

Bottom line

Adding a reverse-curve wire or box elastics to bite turbos cut time to a 30% overbite from 94.15 to 55.72 and 44.42 d (Table 2); the price is 7.46° of proclination (reverse curve) or the largest mandibular-plane opening, +1.84° (elastics). The claimed 1 mm of true intrusion is 0.99 mm at Point I against 0.71 and 0.86 mm without a reverse curve, never tested (Table 4).

Eight numbers to remember

Number	What it is	Say it like this
55.72 ± 10.52 / 44.42 ± 10.88 / 94.15 ± 7.08 d	Days to overbite <30%, I / II / III (Table 2; text prints 55.7 ± 20.5 and 94.4 ± 7.1 d); I vs II 11.3 d, 95% CI 2.32–20.29, P=.010	"Eight, six and thirteen weeks; an auxiliary halves it, and the 11 days between the auxiliaries is under one review interval"
7.46 ± 2.52° / 3.46 ± 3.89° / 1.55 ± 1.39°	Lower incisor to corpus axis, I / II / III (Table 3), P<.001	"The reverse curve flares twice what elastics do and five times turbos alone"
0.99 / 0.71 / 0.86 mm	Point I vertical change, I / II / III (Table 4); no between-group test printed	"Every arm moved Point I about a millimetre, with or without a reverse curve"
8.94 ± 3.45° and 5.99 ± 4.75°	L7 and L6 distal tipping, Group I (Table 3), P<.001 vs II and III	"A cinched reverse curve tips the molars back 6 to 9°; the L7 distal cusp did not extrude (–0.64 mm)"
+1.35 / +1.84 / +0.69°	Mn-Mx change, I / II / III (Table 5); only II vs III differs, P=.011 (FH-Mn P=.026)	"Elastics open the plane most, turbos alone least; the reverse curve differs from neither"
3.92 / 3.65 / 3.57 mm from 5.51 / 5.33 / 5.13 mm	Overbite reduction, I / II / III (Results text; baseline Table 1); no test printed	"Same amount whichever you choose, because the endpoint is a threshold"
81 → 80; 15.82 ± 2.12 / 16.00 ± 2.17 / 17.35 ± 2.92 y	Enrolled, completed (one lost both turbos); mean ages I / II / III (Table 1)	"Adolescents the Discussion calls nongrowing; turbos alone was the oldest arm"
Overbite <30% + Shimstock contact, every 14 d	Correction endpoint and review interval (Methods)	"Duration means time to a threshold, called fortnightly by the unblinded operator"

What each arm did

Arm	How the bite opened	Main side effect	Days
I: turbos + 0.016×0.022 NiTi RCOS, cinched distal to 7s	Incisor tipping and intrusion (tip –2.46 mm; Point I 0.99 mm) plus premolar extrusion 1.05–1.06 mm (Tables 3–4)	Flaring 7.46°, tip 1.01 mm forward; L6/L7 distal tipping 5.99°/8.94°; L4 mesial 4.44°	55.72 ± 10.52
II: turbos + 0.014 NiTi + 3/16-inch medium box elastics	Lower posterior extrusion 0.90–1.19 mm (text: 0.6–1.5 mm) plus upper posterior 0.28–0.42 mm, the only arm with upper extrusion above 0.15 mm (Table 4)	Largest vertical opening: Mn-Mx +1.84°, FH-Mn +1.31°, LFH +1.56° (Table 5); LI-CA +3.46°	44.42 ± 10.88
III: turbos + 0.014 NiTi only	Passive lower posterior eruption 0.86–1.30 mm; incisor tip –1.25 mm; upper incisor –0.43 mm (Table 4)	Slowest; LI-CA +1.55°; Mn-Mx +0.69°	94.15 ± 7.08

Strengths and weaknesses

Strengths

- Prospective; one operator, bracket system and turbo mould; lower arch bonded 7–7.
- Clear endpoint, reviewed every 14 d (monthly in the 2022 RCT, literature).
- Corpus-axis superimposition, Greig's Point I; intraexaminer ICC 0.92–0.98.

Weaknesses

- Operator allocation on alignment and cooperation: Group III oldest (17.35 y), most crowded (1.81 mm, Table 1); Group II compliance-selected.
- Unblinded operator called the endpoint fortnightly, so the 11.3 d gap is inside one visit; no control, no follow-up.
- Standard errors fit arms of about 29/31/20 rather than 27 each, and 14 of 26 turbos-only values match the 2022 RCT's turbo arm (derived; read conflicts.md before raising this).

What it changes at the chair

- An auxiliary cuts bite opening from three months to 6–8 weeks. Caveat: turbos-alone was oldest and most crowded.
- Choose elastics or reverse curve on incisor position and compliance, not the 11.3 d gap (under one visit). Caveat: compliance-selected arm.

- A rectangular NiTi reverse curve as first wire flares incisors $7.46 \pm 2.52^\circ$ even in aligned arches (crowding 1.25 mm). Caveat: Proffit's textbook torque error.
- Not the reverse curve for intrusion: Point I 0.99 vs 0.71 and 0.86 mm without one, untested. Caveat: the 2022 RCT's turbos arm gave 0.28 mm (literature).
- Elastics open the plane most ($+1.84^\circ$) and are the only arm with upper posterior extrusion above 0.15 mm (0.28–0.42 mm). Caveat: long faces excluded; extrusion suits short, growing faces (Proffit).

Three questions to ask

1. "Point I moved 0.99, 0.71 and 0.86 mm, untested between groups; is the millimetre of true intrusion a reverse-curve effect or just turbos on a 0.014 wire?" *Why it lands: their own marker; Group III had no intrusive mechanics.*
2. "Elastics beat the reverse curve by 11 days, CI 2 to 20, with 14-day reviews and a compliance-selected arm; is that gap real, or do you choose on 3.5° versus 7.5° of flaring?" *Why it lands: Tables 2 and 3 together; gap under one visit.*
3. "Turbos alone extruded the lower posteriors as much as elastics, 1.30 versus 1.12 mm at the first-molar mesial cusp, yet took twice as long; where does the speed come from?" *Why it lands: Table 4 read across arms; only Group II extruded upper posteriors beyond 0.15 mm (0.28–0.42 mm).*

Five questions you will be asked

- **Primary outcome?** Days from bonding to overbite <30% with Shimstock contact: 55.72/44.42/94.15 d; undeclared, unregistered.
- **Baseline balance?** No $P<.05$ (Table 1), but Group III oldest (17.35 y), most crowded (1.81 mm); Group I ANB 5.14° despite a 5° ceiling.
- **Reverse-curve effects?** LI-CA $+7.46^\circ$, tip -2.46 mm, 1.01 mm forward, Point I 0.99 mm; L6/L7 $+5.99^\circ/+8.94^\circ$ distal, L4 $+4.44^\circ$ mesial.
- **Who gets elastics?** A compliant, short- or average-faced adolescent with forward incisors: 44.42 d, $+3.46^\circ$; not long faces, Mn-Mx $+1.84^\circ$.
- **Evidence level; practice change?** CEBM Level 3. Add an auxiliary when speed matters, chosen on inclination and compliance; claim neither intrusion nor stability.

Exam radar

Ten facts most likely to be tested

1. Non-randomised; operator allocated on alignment, inclination, cooperation.
2. Durations (Table 2): 55.72/44.42/94.15 d, all pairs significant; I-II 11.3 d, $P=.010$.
3. Endpoint: overbite <30% plus Shimstock, fortnightly; reduction 3.92/3.65/3.57 mm from 5.51/5.33/5.13 mm, untested.
4. LI-CA (Table 3): $+7.46^\circ/+3.46^\circ/+1.55^\circ$, $P<.001$.
5. Superimposition: corpus axis at Pm (Ricketts), palatal vault (Broadbent); Point I at $0.66 \times$ incisor length from the tip (Greig): 0.99/0.71/0.86 mm, untested.
6. Group I (Table 3): L6 5.99° , L7 8.94° distal; L4 4.44° mesial.
7. Group II (Table 4): lower extrusion 0.90–1.19 mm; upper 0.28–0.42 mm (other arms ≤ 0.15 mm).
8. Mn-Mx (Table 5): $+1.35^\circ/+1.84^\circ/+0.69^\circ$; only II-III differ ($P=.011$; FH-Mn $P=.026$).
9. Ages 15.82/16.00/17.35 y, guardian consent, called nongrowing; no untreated control.
10. Inclusion: overbite >40%, ANB 1–5°, average/reduced LFH, ≤ 3 mm crowding, average smile; overjet >6 mm excluded.

Three likely MCQ stems

1. Fastest arm and margin over the reverse curve? A. Reverse curve, 11 d. B. Elastics, 11 d. C. Turbos alone, 39 d. D. Elastics, 50 d. **B:** Table 2, 44.42 vs 55.72 d.
2. Proclined lower incisors, 55% overbite: which protocol adds least proclination? A. Turbos+RCOS. B. Turbos+elastics. C. Turbos+0.014 NiTi. D. All equal. **C:** LI-CA $+1.55^\circ$ (Table 3).
3. The Conclusions credit the reverse curve with 1 mm of true intrusion. What does Table 4 actually show? A. Only Group I intruded. B. Similar Point I change in all arms, never tested between groups. C. 0.014 NiTi intrudes as well as a reverse curve. D. A group-size artefact. **B:** 0.99/0.71/0.86 mm, untested.

One likely oral question

"17-year-old woman, Class I, 60% overbite, average face, upright incisors, 2 mm crowding, wants speed: which protocol, what warnings?"

She hits the criteria; turbos with box elastics: 44.42 d against 55.72 and 94.15 d (Table 2). Warnings: elastics work only when worn; posteriors extrude about 1 mm; the mandibular plane opens 1.8° (Tables 4–5). Upright incisors could take the reverse curve's 7.46° , but elastics get there with 3.46° and no molar tip-back (Table 3). No follow-up past a 30% overbite: I promise neither total time nor stability; Proffit's bite-plane retainer for years applies.

Before the session

When	Do	Time
Day received	Read summary.md and key-numbers.md; mark the eight numbers	40 min
Next day	Run the flashcards; rehearse the five answers aloud with the numbers	30 min
Day before	This handout only	20 min
In the room	Open with the 30-second version; ask question 1 first	—

Verdict in one line

A reverse-curve wire or box elastics on bite turbos halves time to a 30% overbite in short- or average-faced, well-aligned adolescents; the ranking between them, the millimetre of intrusion and every skeletal contrast sit inside operator-allocation noise.